

# 1 Mean-variance and Optimization

## 1.1 Portfolio Constraints

**Review: Markowitz Mean-Variance Problem**

Recall the standard Markowitz Mean-Variance Problem:

$$\begin{aligned} \min_{w_0, w} \quad & \frac{1}{2} w^\top C w \\ \text{s.t.} \quad & w_0 r_f + w^\top \mu = z \\ & w_0 + w^\top \mathbf{1}_n = 1 \end{aligned}$$

where  $w_0$  is the weight in the risk-free asset,  $w$  is the vector of weights in risky assets,  $C$  is the covariance matrix, and  $\mu$  is the expected return vector.

**Equivalent formulation:**

$$\begin{aligned} \min_w \quad & \frac{1}{2} w^\top C w \\ \text{s.t.} \quad & w^\top (\mu - r_f \mathbf{1}_n) = z - r_f \end{aligned}$$

The above formulation does not take into accounts of constraints in real trading.

### Basic Trading Constraints

**No Short selling:**  $w_i \geq 0$ , for  $i = 1, \dots, n$ .

**No borrowing:**  $w_0 \geq 0$ , which implies  $\sum_{i=1}^n w_i \leq 1$ .

### Margin Requirement for Short Sale

Suppose that for stock  $i$ , the short sale of account of this stock should have  $(1 + m_1)$  of the value of the short sale. Suppose the margin account earns a deterministic net return rate  $r_c$ . Denote by  $w_i$  the amount of margin prepared for the short sale of stock  $i$ , and denote  $v_i := (w_1, \dots, w_n)$ .

**Portfolio Net Return:**

$$\begin{aligned} R_{(w,v)} &= w^\top R + v^\top r_c \mathbf{1}_n + (1 - w^\top \mathbf{1}_n - v^\top \mathbf{1}_n) r_f \\ \Rightarrow R_{(w,v)} &= r_f + w^\top (R - r_f \mathbf{1}_n) + v^\top (r_c - r_f) \mathbf{1}_n \end{aligned}$$

**Moments:** Variance is given by  $\text{Var}(R_{(w,v)}) = w^\top C w$ . Expectation is given by

$$\mathbb{E}[R_{(w,v)}] = r_f + w^\top (\mu - r_f \mathbf{1}_n) + v^\top (r_c - r_f) \mathbf{1}_n$$

**Constraints:** Constraints to model margin requirements:

$$v_i \geq -(1 + m_1) w_i, \quad v_i \geq 0, \quad i = 1, \dots, n.$$

### Cardinality Constraints

Due to various forms of market friction, most investors can invest in only a limited number of assets, leading to cardinality constraints. Suppose one can invest in at most  $s$  stocks. The cardinality constraint can be formulated as the following linear constraints:

$$\begin{aligned} -M \delta_i \leq w_i \leq M \delta_i, \quad \delta_i \in \{0, 1\}, \quad i = 1, \dots, n \\ \sum_{i=1}^n \delta_i \leq s \end{aligned}$$

where  $M$  is a given, sufficiently large number, and  $\delta_i$  denotes whether one invests in stock  $i$ .

### Buy-In Threshold Constraints

Small positions have very limited impact on the total performance of the portfolio, but trigger non-negligible establishing and maintaining cost. To avoid small positions, one can impose a buy-in threshold constraint to prevent from holding an active position below a prescribed proportion  $w_{\min} > 0$  of the total capital. The buy-in threshold constraint can be modeled as the following linear constraints:

$$w_i \leq M \delta_i, \quad w_{\min} \delta_i \leq w_i, \quad \delta_i \in \{0, 1\}, \quad i = 1, \dots, n$$

where  $M$  is a given, sufficiently large number.

### Diversification Constraints

Suppose that the  $n$  stocks are divided into  $K$  sectors, and  $S_k$  is the collection of the identities of stocks in section  $k$ ,  $k = 1, \dots, K$ . Suppose that the investor wants to invest in at least  $K_{\min}$  sectors, and for each sector she wants to invest, the minimum predefined level of investment is  $s_{\min}$ . The above diversification constraint can be modeled as the following linear constraints:

$$\sum_{k=1}^K s_k \leq \sum_{i \in S_k} w_i \leq M \delta_k, \quad k = 1, \dots, K$$

where  $M$  is a given, sufficiently large number.

### Proportional Transaction Costs and Bid-Ask Spread

Suppose that there is a proportional transaction cost that amounts to  $c_{v,b}$  dollars for each dollar invested in a stock, and a similar cost  $c_{v,s}$  when selling the stock at the end of the period. For simplicity, suppose short sales are not allowed, and let  $w_i$  represent the proportion of the initial capital, pre-transaction cost, invested in stock  $i$ . Proportional transaction costs can be taken into account by modifying the net return rate of a stock  $R_i$ , which is computed from the stock price directly, as follows:

$$R'_i := \frac{1 - c_{v,s}}{1 + c_{v,b}} (1 + R_i) - 1$$

To take into account the bid-ask spread, one only needs to use the ask price at the beginning of the period and the bid price at the end of the period to compute the net return rate  $R_i$ .

## 1.2 Relative Entropy Model

Suppose that the agent is uncertain about the model, which is represented by the probability density function of the excess return of the risky assets. The agent has a reference model  $f_0$  but considers alternative models  $f$ . Consider the log likelihood ratio of an alternative model  $f$  and the reference model  $f_0$  based on observed asset returns  $X_t$ :

$$R(X_1, \dots, X_T) = \sum_{t=1}^T \log(f(X_t)/f_0(X_t))$$

By the law of large numbers, under the model  $f$ :

$$\lim_{T \rightarrow \infty} \frac{1}{T} R(X_1, \dots, X_T) = \mathbb{E}[\log(f(X_t)/f_0(X_t))] = \mathbb{E}^0[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))]$$

The term  $\mathbb{E}^0[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))]$  is called the relative entropy of the model  $f$  with respect to the reference model  $f_0$ .

The agent faces the following problem:

$$\min_{f \in \mathcal{F}} \max_w \lambda w^\top \mathbb{E}[X] - \frac{1}{2} w^\top \text{Var}^f(X) w$$

where  $X$  stands for the excess return vector of the risky assets and  $\text{Var}^f$  stands for the variance under  $f$ , and

$$\mathcal{F} := \{f \mid \mathbb{E}[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))] \leq \eta\}$$

for some positive parameter  $\eta$ . Assume that the excess return vector follows joint normal distribution. Assume that the covariance matrix  $C$  is known, and alternative model  $f$  differs from the reference model  $f_0$  in the mean. Denote the mean under  $f$  and  $f_0$  as  $\mu$  and  $\mu_0$ , respectively. Then, the relative entropy of  $f$  is:

$$\mathbb{E}^0[\log(f(X_t)/f_0(X_t)) \log(f(X_t)/f_0(X_t))] = \frac{1}{2} (\mu - \mu_0)^\top C^{-1} (\mu - \mu_0)$$

The agent's problem becomes:

$$\min_{\mu} (\mu - \mu_0)^\top C^{-1} (\mu - \mu_0) \leq 2\eta \max_w \lambda w^\top \mu - \frac{1}{2} w^\top C w$$

**Optimal solution:**

If  $\mu_0^\top C^{-1} \mu_0 \leq 2\eta$ , then  $w^* = 0$ .

$$\text{If } \mu_0^\top C^{-1} \mu_0 > 2\eta, \text{ then } w^* = \lambda \left( 1 - \sqrt{\frac{2\eta}{\mu_0^\top C^{-1} \mu_0}} \right)^\top C^{-1} \mu_0$$

## 1.3 A Smooth-Ambiguity Model

The worst-case mean-variance portfolio selection is usually too conservative. We consider the smooth-ambiguity model, in which the investor has a belief of the model/parameters for the stock distributions, considers the average model/parameter setup instead of the worst-case scenario, and is averse to ambiguity about the model/parameters.

Formally, denote by  $P = (\mu, C)$  a model specification. The investor has a belief of  $P$ , represented by distribution  $\pi$ . Let  $\bar{P} = (\bar{\mu}, \bar{C}) = \mathbb{E}_\pi[P]$  be the average model specification. The portfolio selection can be formulated as:

$$\max_{w_0} \mathbb{E}_\pi[R_w] - \frac{\lambda}{2} \text{Var}_\pi(R_w) - \frac{\theta}{2} \text{Var}_\pi(\mathbb{E}_P[R_w])$$

where  $\lambda > 0$  is the risk aversion parameter and  $\theta > 0$  is the ambiguity aversion parameter.

# 2 Mean-Risk Opt and EU Max

Denote by  $R = (R_1, \dots, R_n)^\top$  the net return vector of the stocks and  $r_f$  the risk-free net return rate. Given portfolio  $w = (w_1, \dots, w_n)^\top$ , where  $w_i$  represents the percentage of initial wealth invested in stock  $i$ , the net return rate of the portfolio is:

$$R_w = r_f + w^\top (R - r_f \mathbf{1}_n)$$

## 2.1 Mean-Semi-Variance Model

One alternative to variance is semi-variance:

$$\mathbb{E}[(\min\{R_w - \mathbb{E}[R_w], 0\})^2]$$

This measures the downside risk of the portfolio. If the distribution is symmetrical, semi-variance is proportional to variance; in general, it is not the case. The Mean-semi-variance portfolio selection problem is formulated as:

$$\begin{aligned} \min_{w_0} \mathbb{E}[(\min\{R_w - \mathbb{E}[R_w], 0\})^2] \\ \text{s.t.} \quad \mathbb{E}[R_w] \geq z \end{aligned}$$

## 2.2 Mean-Absolute Deviation Model

Another alternative to variance is absolute deviation:

$$\mathbb{E}[|R_w - \mathbb{E}[R_w]|]$$

Mean-absolute deviation portfolio selection is formulated as:

$$\begin{aligned} \min_{w_0} \mathbb{E}[|R_w - \mathbb{E}[R_w]|] \\ \text{s.t.} \quad \mathbb{E}[R_w] \geq z \end{aligned}$$

## 2.3 Safety-First Model

Safety-first portfolio selection is formulated as:

$$\begin{aligned} \min_{w_0} \mathbb{P}(R_w \leq D) \\ \text{s.t.} \quad \mathbb{E}[R_w] \geq z \end{aligned}$$

Here,  $D$  is a given threshold below which the portfolio return is considered to be a disaster. Both the mean-semi-variance and the mean-absolute deviation portfolio selection problems are convex programs, but the safety-first portfolio selection problem is not.

## 2.4 Tail Risk Measures

In the safety-first criterion, investors are concerned about the left tail of the return distribution. Given initial wealth  $x_0$  and a portfolio  $w$ , the wealth at the end of the period is  $X_T = x_0(1 + R_w)$ , and thus the P&L of the portfolio is:

$$Z := X_T - x_0 = x_0 R_w$$

### Definition of VaR

Let  $Z$  be a random P&L, with negative values referring to losses and positive values to gains. Given a level  $\alpha \in (0, 1)$ , informally the  $\alpha$ -VaR, denoted as  $\text{VaR}_\alpha(Z)$ , is defined by:

$$\mathbb{P}(-Z \leq \text{VaR}_\alpha(Z)) = \alpha$$

Formally,  $\text{VaR}_\alpha(Z)$  is defined as follows:

$$\begin{aligned} \text{VaR}_\alpha(Z) &= \inf\{x \in \mathbb{R} : \mathbb{P}(-Z \leq x) \geq \alpha\} \\ &= -\sup\{y \in \mathbb{R} : \mathbb{P}(Z < y) \leq 1 - \alpha\} \end{aligned}$$

### Definition of CVaR

Given a level  $\alpha \in (0, 1)$ , informally the  $\alpha$ -CVaR, denoted as  $\text{CVaR}_\alpha(Z)$ , is defined by:

$$\text{CVaR}_\alpha(Z) = \mathbb{E}[-Z] - Z \geq \text{VaR}_\alpha(Z)$$

Formally,  $\text{CVaR}_\alpha(Z)$  is defined by:

$$\text{CVaR}_\alpha(Z) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_\beta(Z) d\beta$$

CVaR can be derived by solving an optimization problem:

$$\text{CVaR}_\alpha(Z) = \min_b \{b + \frac{1}{1 - \alpha} \mathbb{E}[\max(-Z - b, 0)]\}$$

## 2.5 Coherent Risk Measures

A risk measure  $\rho(Z)$  assigns a numerical value to a random P&L  $Z$ . A risk measure  $\rho$  is a coherent risk measure, if it satisfies the following properties:

**Monotonicity:** for any  $Z_1$  and  $Z_2$  with  $Z_1 \leq Z_2$ ,  $\rho(Z_1) \geq \rho(Z_2)$ .

**Translation invariance:** for any constant  $m$  and any  $Z$ ,  $\rho(Z + m) = \rho(Z) - m$ .

**Positive homogeneity:** for any  $Z$  and any positive constant  $\lambda$ ,  $\rho(\lambda Z) = \lambda \rho(Z)$ .

**Subadditivity:** for any P&Ls  $Z_1$  and  $Z_2$ ,  $\rho(Z_1 + Z_2) \leq \rho(Z_1) + \rho(Z_2)$ .

CVaR is a coherent risk measure. VaR satisfies monotonicity, translation invariance, and positive homogeneity, but not subadditivity; thus, it is not a coherent risk measure.

## 2.6 Data-Driven Mean-Risk Portfolio Selection

Suppose that we have observed  $T$  samples of  $R$ , denoted as  $R_t$ ,  $t = 1, \dots, T$ . Assume that the future scenarios are confined to be within these  $T$  observed samples, and each has a probability of  $1/T$  to occur. This means the distribution of the return rate in the next period is the empirical distribution. Denote the sample mean as:

$$\bar{R} := \frac{1}{T} \sum_{t=1}^T R_t$$

### Optimization: Mean-Semi-Variance Portfolio Selection

The data-driven mean-semi-variance portfolio selection can be formulated as the following quadratic program(no borrowing and no shorting.):

$$\begin{aligned} \min_{w_0, w_1, \dots, w_T} \quad & \frac{1}{T} \sum_{t=1}^T w_t^2 \\ \text{s.t.} \quad & y_t \geq w_t^\top (\bar{R} - R_t), \quad t = 1, \dots, T \\ & y_t \geq 0, \quad t = 1, \dots, T \\ & r_f + w^\top (\bar{R} - r_f \mathbf{1}_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1, \dots, n \end{aligned}$$

### Optimization: Mean-Absolute Standard Deviation Portfolio Selection

The data-driven mean-absolute standard deviation portfolio selection can be formulated as the following linear program(no borrowing and no shorting):

$$\begin{aligned} \min_{w_0, w_1, \dots, w_T} \quad & \frac{1}{T} \sum_{t=1}^T y_t \\ \text{s.t.} \quad & y_t \geq w_t^\top (\bar{R} - R_t), \quad t = 1, \dots, T \\ & y_t \geq -w_t^\top (R_t - \bar{R}), \quad t = 1, \dots, T \\ & r_f + w^\top (\bar{R} - r_f \mathbf{1}_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1, \dots, n \end{aligned}$$

### Optimization: Mean-CVaR Portfolio Selection

Suppose the initial wealth is  $x_0$ , so for portfolio  $w$ , the P&L is  $Z_w = x_0 R_w$ . Suppose the target for the expected return rate is  $z$ . The data-driven mean- $\alpha$ -CVaR portfolio selection can be formulated as the following linear program:

$$\begin{aligned} \min_{w_0, w_1, \dots, w_T} \quad & b + \frac{1}{(1 - \alpha)T} \sum_{t=1}^T y_t \\ \text{s.t.} \quad & y_t \geq -(r_f + w^\top (\bar{R} - r_f \mathbf{1}_n)) x_0 - b, \quad t = 1, \dots, T \\ & y_t \geq 0, \quad t = 1, \dots, T \\ & r_f + w^\top (\bar{R} - r_f \mathbf{1}_n) \geq z \\ & 0 \leq w_i \leq 1, \quad i = 1, \dots, n \end{aligned}$$

## 2.7 Expected Utility

Expected utility (EU) theory suggests that when making decisions under risk, people do not evaluate monetary payoffs directly. Instead, they attribute a subjective value (utility) to each monetary payoff.  $X$  is evaluated as the expectation of its utility  $\mathbb{E}[u(X)]$ . With a logarithmic utility  $u(x) = \log(x)$ , the expected utility of the St. Petersburg game is:

$$\mathbb{E}[u(X)] = \sum_{n=1}^{\infty} (1/2^n) \times \log(2^n) = \log 2 \times \sum_{n=1}^{\infty} n/2^n = \log(4)$$

This solves the paradox.

### Risk Aversion

Individuals are risk averse if any random payoff  $X$  is less preferred to its mean  $\mathbb{E}[X]$ . An individual with EU preferences is risk averse if and only if their utility function  $u$  is concave. Concavity is equivalent to diminishing sensitivity, meaning the marginal utility  $u'(x)$  is decreasing.

### Certainty Equivalent

The certainty equivalent of a random payoff  $X$  is the deterministic payoff  $c$  that is indifferent to  $X$ :

$$u(c) = \mathbb{E}[u(X)]$$

Individual  $A$  is more risk averse than  $B$  if the certainty equivalent of  $X$  for  $A$  is less than or equal to that for  $B$ .

### Risk Aversion Coefficient

The Arrow-Pratt absolute risk aversion coefficient is defined as:

$$-\frac{u''(x)}{u'(x)}$$

The larger the coefficient, the more concave the utility function and the more risk averse the individual.

### Equivalent Utility Functions

Preferences remain the same if we replace  $u(x)$  with  $au(x) + b$  where  $a > 0$ . Thus,  $v(x) := au(x) + b$  is equivalent to  $u(x)$ .

### Examples of Utility Functions

Exponential utility ( $\gamma > 0$ ):

$$u(x) = 1 - e^{-\gamma x}, \quad x \in \mathbb{R}$$

Power utility ( $\gamma > 0, \gamma \neq 1$ ):

$$u(x) = x^{1-\gamma}/(1-\gamma), \quad x > 0$$

Logarithmic utility:

$$u(x) = \ln x, \quad x > 0$$

Quadratic utility ( $\gamma > 0$ ):

$$u(x) = x - \frac{\gamma}{2} x^2, \quad x \in \mathbb{R}$$

## 2.8 Elicit Utility Function

### Certainty Equivalent Method

Select reference points  $A$  and  $B$ . Propose lottery  $(A, p; B, 1 - p)$ . Ask for certainty equivalent  $c$ . Then:

$$u(c) = pu(A) + (1 - p)u(B)$$

$u(A)$  and  $u(B)$  are set exogenously due to scaling freedom. Varying  $p$  maps the curve.

### Probability Equivalent Method

Select reference points  $A$  and  $B$ . For a value  $x$ , ask for the probability  $p$  such that the individual accepts the lottery in place of  $x$ . Then:

$$u(x) = pu(A) + (1 - p)u(B)$$

### Tradeoff Method

Compare lotteries  $(X, p; r)$  and  $(x, p; r)$  with reference outcomes  $R > r$ . Vary  $X$  until indifference. Then compare  $(Y, p; r)$  and  $(y, p; r)$ . By EUT:

$$u(X) - u(x) = u(Y) - u(y)$$

This establishes an indifference relation. Starting from a base  $x_0$ , one can identify a sequence where  $u(x_j) = ju(x_1)$ .

## 2.9 Portfolio Selection with EU in Single Period

Consider a single period problem with initial wealth  $x_0$ , assets with returns  $R_t$ , and portfolio weights  $w_t$ . Portfolio return is  $R_w = \sum_{i=1}^n w_i R_i$ . Terminal wealth is  $x_0(1 + R_w)$ . The objective is:

$$\max_{w_t} \mathbb{E}[u(x_0(1 + R_w))] \\ \text{s.t.} \quad \sum_{i=1}^n w_i = 1$$

### Relation with Mean-Variance Criterion

For quadratic utility  $u(x) = x - (\gamma/2)x^2$ , the expected utility depends on  $\mathbb{E}[R_w]$  and  $\text{Var}[R_w]$ . Thus, the optimal portfolio must be in the mean-variance set.

Expected utility of terminal wealth:

$$\begin{aligned} x_0 \left\{ \mathbb{E}[1 + R_w] - (\gamma x_0/2) \mathbb{E}[(1 + R_w)^2] \right\} \\ = x_0 \left\{ 1 - (\gamma x_0/2) + (1 - \gamma x_0) \mathbb{E}[R_w] + (\gamma x_0/2) (\mathbb{E}[R_w])^2 - (\gamma x_0/2) \text{Var}[R_w] \right\} \end{aligned}$$

For exponential utility  $u(x) = -e^{-\gamma x}$  and normally distributed returns, the expected utility is:

$$-\exp(-\gamma x_0(1 + \mathbb{E}[R_w]) + \frac{\gamma^2 x_0^2}{2} \text{Var}[R_w])$$

Thus, the optimal portfolio must be on the mean-variance efficient frontier.

### General Case

In general, EU and mean-variance criteria differ. Using the Lagrange dual method, the optimal portfolio  $w^*$  satisfies:

$$\begin{aligned} \mathbb{E}[u'(x_0(1 + R_w))] x_0 R_i &= \lambda, \quad i = 1, \dots, n \\ \text{s.t.} \quad \sum_{i=1}^n w_i^* &= 1 \end{aligned}$$

# 3 CAPM and Arbitrage Pricing Theory

## 3.1 Market Equilibrium and CAPM model

**Market equilibrium:** at equilibrium price

All investors have achieved their goals. Market is clear, demand=supply.

**Capital asset pricing model (CAPM):** an equilibrium pricing model in framework of mean-variance analysis.

### Assumptions

Each investor is a price-taker and mean-variance optimizer (although with different risk appetite). Everyone agrees on the probabilistic structure of assets (mean, covariance matrix of future asset prices). There is a unique risk-free rate  $r_f$  of borrowing and lending that is available to all. Market is frictionless: assets can be traded in any fraction and there is no transaction fee and other fees. The equilibrium price exists.

### Market Portfolio

Suppose there are  $n$  risky assets in the market with number of shares  $N_i$ ,  $i = 1, \dots, n$ . The current prices are  $P_i$ ,  $i = 1, \dots, n$ .

Total capital:  $\sum_{i=1}^n N_i P_i$

Capitalization weight:  $\tilde{w}_M = (\tilde{w}_M, \dots, \tilde{w}_M, n)$

$$\tilde{w}_{M,j} = \frac{\text{capital in asset } j}{\text{total capital}} = \frac{N_j P_j}{\sum_{i=1}^n N_i P_i}, \quad i = 1, \dots, n$$

Market portfolio: portfolio with capitalization weight.

### Market Portfolio and Tangent Portfolio

### Arbitrage Pricing Theory (APT)

Theorem (Arbitrage Pricing Theory (APT)): Assume there is no arbitrage opportunity in the market. Then,  $\alpha = 0$ . In consequence,

$$\mathbb{E}[r] = \beta \mathbb{E}[r_f]$$

$\mathbb{E}[r]$  is the risk premium of the common factors. The risk premium of asset  $i$  is its loading on the common factors multiplied by the risk premia of the common factors.  
In CAPM, the only factor is the excess return of the market portfolio,  $R_M - r_f$ . The risk premium of the common factor is the expected excess return of the market portfolio,  $\mathbb{E}[R_M] - r_f$ . The loading of asset  $i$  on the common factor is its beta.

#### Statistical Testing of APT

Suppose we have historical data of the payoffs of the  $n$  zero-cost strategies:  $\tilde{r}_i$ ,  $t = 1, \dots, T$  and the historical data of the payoffs of the  $m$  factors:  $f_{it}$ ,  $t = 1, \dots, T$ . Suppose the payoffs are i.i.d. across periods. Run multiple linear regression to estimate  $\alpha$  and  $\beta$ . Reject APT if the estimate of  $\alpha$  is significantly different from zero.

## 4 Multi-Period Investment Models

### 4.1 Multi-Period Investment

There are  $N$  periods of investment. Denote the beginning and end of  $j$ -th period as  $t_{j-1}$  and  $t_j$ ,  $j = 1, \dots, N$ . There is a risk-free asset and  $m$  risky stocks in the market. The net return rate of investing in the risk-free asset in the  $j$ -th period is a constant  $r_{f,j}$ . The excess return rate of investing in stock  $i$  in period  $j$  is  $R_{i,j}$ .

Assume that  $R_j := (R_{1,j}, \dots, R_{m,j})^\top$ ,  $j = 1, \dots, N$  are independent.

Consider an agent with initial wealth  $w_0$  at the beginning of first period. The agent's decision is the allocation of his money in each period.

### 4.2 Wealth Dynamics

Denote as  $X_j$  the wealth of the agent at time  $t_j$ ,  $j = 0, 1, \dots, N$ . Suppose that except for the initial investment, the agent does not inject new money nor withdraw money from the investment account before the terminal time  $t := t_N$ . At the beginning of each period  $j$ , the agent decides to invest  $\pi_{j-1}$  dollars in stock  $i$ ,  $i = 1, \dots, m$ , and the remaining amount of money in the risk-free asset. Wealth  $X_j$  is

$$X_j = (1 + r_{f,j})X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

### 4.3 Expected Utility Maximization

Assume that the agent maximizes the expected utility of her terminal wealth,

$$\max_{\pi_{j-1}, j=1, \dots, N} \mathbb{E}[u(X_N)]$$

where  $u$  is the agent's utility function and  $\pi_{j-1} = (\pi_{1,j-1}, \dots, \pi_{m,j-1})^\top$ ,  $j = 1, \dots, N$  are the agent's portfolio strategy.

### 4.4 Dynamic Programming

We apply dynamic programming to solve the problem. Principle of Optimality: Starting from any point on the optimal path, the remaining decisions must constitute an optimal policy with regard to the point.

Consider time  $t_k$  when the agent's wealth then is  $x$ . Consider the following problem:

$$\max_{\pi_{j-1}, j=k+1, \dots, N} \mathbb{E}_k[u(X_N) | X_k = x]$$

Denote the above optimal value function as  $v_k(x)$ .

Dynamic Programming (DP) Equation:

$$v_k(x) = \max_{\pi_k} \mathbb{E}_k[v_{k+1}(X_{k+1}) | X_k = x], \quad k = 0, 1, \dots, N-1$$

$$v_N(x) = u(x)$$

Instead of solving a dynamic portfolio selection problem, we solve a family of single-period portfolio selection problem.  $(\pi_0^*, \dots, \pi_{N-1}^*)$  is optimal to the dynamic portfolio selection problem if and only if  $\pi_k^*$  is optimal to the above single-period portfolio selection problem,  $k = 0, \dots, N-1$ .

### 4.5 Power Utility

In the following, we assume power utility function:

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad x > 0$$

for some  $0 < \gamma \neq 1$ . Applying DP, we have:

$$v_{N-1}(x) = \max_{\pi_{N-1}} \mathbb{E}_{N-1}[u(X_N) | X_{N-1} = x] = b_{N-1} x^{1-\gamma}, \quad x > 0$$

where

$$b_{N-1} = \max_w \frac{\mathbb{E}[(1+r_f+w^\top R_N)^{1-\gamma}]}{1-\gamma}$$

Denote the optimizer in the right-hand side of the above as  $w_{N-1}^*$ .

By induction, we conclude:

$$v_{k-1}(x) = \left( \prod_{j=k}^{N-1} (b_j(1-\gamma)) \right) \frac{x^{1-\gamma}}{1-\gamma}$$

where

$$b_k = \max_w \frac{\mathbb{E}[(1+r_f+w^\top R_k)^{1-\gamma}]}{1-\gamma}, \quad k = 1, \dots, N$$

Denote the optimizer in the right-hand side of the above as  $w_{k-1}^*$ . Then, the optimal portfolio is:

$$\pi_{k-1}^* = w_{k-1}^* X_{k-1}, \quad k = 1, \dots, N$$

#### Myopic Portfolio

With the power utility function, the investor acts at each stage, based on the optimal policy, as if he were faced with a single-period investment, i.e., as if he were myopic. When the utility function is not a power function or when the stock returns are not independent over time, the above is not necessarily true. For power utility functions, it is implicitly assumed that the wealth must remain positive. Thus, the portfolio  $\pi$  needs to be constrained, i.e.,  $\pi_{i,k} \geq 0$ ,  $i = 1, \dots, n$ , and  $\sum_{i=1}^n \pi_{i,k} \leq X_k$ .

#### Example: Expected Utility Maximization

Suppose the risk-free net return rate in each period is 0.03, and there is one stock in the market only with i.i.d. excess return rates over time. Suppose that the log return rate of the stock,  $\ln(1+r_f+R_k)$  follows normal distribution with mean 0.07 and variance 0.04.

Suppose that the investor is not allowed to short-sell the stock or to borrow, i.e.,  $0 \leq \pi_k \leq X_k$ . Suppose that the investor has a power utility function with  $\gamma = 10$ . To find the optimal portfolio, we need to solve:

$$\max_{w \in [0,1]} f(w)$$

where

$$f(w) := \frac{\mathbb{E}[(1+r_f+R_k w)^{1-\gamma}]}{1-\gamma}$$

Straightforward calculation yields:

$$f'(w) = \mathbb{E}[(1+r_f+R_k w)^{-\gamma} R_k]$$

By binary search, one can find that  $f'(0.1489) = 0$ , which implies that the optimal portfolio  $w^* = 0.1489$ , i.e., it is optimal to invest 14.89

### 4.6 Long-Run Mean Exponential Growth Rate

Suppose  $t_k - t_{k-1} = \Delta$ ,  $k = 1, \dots, N$  in the above, e.g.,  $\Delta$  is one month, so the terminal time  $t_N = N\Delta$ . Assume that the stock excess returns  $R_k$ ,  $k = 1, 2, \dots$  are i.i.d. For each portfolio strategy  $(\pi_1, \pi_2, \dots)$ , consider the long-run mean exponential growth rate:

$$\lim_{N \rightarrow +\infty} \mathbb{E}_1 \left[ \frac{\log(X_N/X_0)}{N} \right]$$

The objective is to maximize the above long-run mean exponential growth rate.

#### Kelly's Portfolio

It turns out that the optimal strategy is as follows: at each time  $k\Delta$  invest  $w_k^\top$  proportion of wealth at that time in stock  $i$ ,  $i = 1, \dots, n$ , where

$$w^* := (w_1^*, \dots, w_n^*)^\top = \arg \max_{w \geq 0} \mathbb{E}[\log(1+r_f+w^\top R_k)]$$

The above strategy is called Kelly strategy.

Kelly portfolio has many interesting properties: Asymptotically maximizes the rate of asset growth. Asymptotically maximizes the median of wealth. Is a myopic strategy.

The Kelly strategy may lead to a large fraction of wealth in bets or in risky investment. For instance, for a stock with annual mean excess return rate 4% and volatility 20%, the Kelly strategy is to invest 100% of wealth in the stock.

#### Fractional Kelly

MacLean et al. (1992) propose fractional Kelly to tradeoff growth and security.  $\kappa$ -fractional Kelly refers to a strategy of investing  $\kappa$  fraction of one's wealth in the Kelly portfolio and holding the remaining as cash (or in the risk-free asset). Fractional Kelly is a heuristic portfolio, rather than the consequence of optimality. MacLean et al. (1992) lay out two attractive properties of fractional Kelly: It is easily computable. There is an effective tradeoff between growth and security by varying  $\kappa$ .

### 4.7 Multi-Period Mean-Variance Portfolio

Next, we consider multi-period mean-variance portfolio selection. The market efficiency is as before. Denote by  $\mu_k$  and  $C_k$  the mean and variance of  $R_k$ , respectively.

At time 0, given initial wealth  $w_0$ , the agent considers the following dynamic portfolio selection problem:

$$\min_{\pi_0, \dots, \pi_{N-1}} \frac{1}{2} \text{Var}(X_N)$$

$$\text{s.t. } \mathbb{E}[X_N] = z_0$$

$$X_j = (1 + r_{f,j})X_{j-1} + \pi_{j-1}^\top R_j, \quad j = 1, \dots, N$$

$$X_0 = w_0$$

where  $z_0$  is the agent's target for expected terminal wealth.

#### Expected Quadratic Utility

Assume that  $C_k$  is positive definite,  $k = 1, \dots, N$ .

Assume that  $z_0 > \prod_{j=1}^N (1 + r_{f,j})$ .

The mean-variance portfolio selection problem is equivalent to the following:

$$\min_{\pi_0, \dots, \pi_{N-1}} \frac{1}{2} \mathbb{E}[(X_N - z_0)^2]$$

$$\text{s.t. } \mathbb{E}[X_N] = z_0$$

$$X_j = (1 + r_{f,j})X_{j-1} + \pi_{j-1}^\top R_j, \quad j = 1, \dots, N$$

$$X_0 = w_0$$

#### Lagrange Dual Method

Apply the Lagrange multiplier  $\lambda_0$  to the constraint for the expected terminal wealth target, and we obtain:

$$\min_{\pi_0, \dots, \pi_{N-1}} \frac{1}{2} \mathbb{E}[(X_N - z_0)^2] - \lambda_0 (\mathbb{E}[X_N] - z_0)$$

subject to

$$X_j = (1 + r_{f,j})X_{j-1} + \pi_{j-1}^\top R_j, \quad j = 1, \dots, N$$

$$X_0 = w_0$$

Then, we apply dynamic programming to solve the above problem: for each  $k = 0, 1, \dots, N-1$  and each  $x$ , consider:

$$\min_{\pi_k, \dots, \pi_{N-1}} \frac{1}{2} \mathbb{E}_k[(X_N - z_0)^2 | X_k = x] - \lambda_0 (\mathbb{E}_k[X_N | X_k = x] - z_0)$$

$$\text{s.t. } X_j = (1 + r_{f,j})X_{j-1} + \pi_{j-1}^\top R_j, \quad j = k+1, \dots, N$$

$$X_k = x$$

Denote the optimal value of the above as  $v_k(x)$ .

#### Dynamic Programming

DP principle:

$$v_k(x) = \min_{\pi_k} \mathbb{E}_k[v_{k+1}((1 + r_{f,k+1})X_k + \pi_k^\top R_{k+1}) | X_k = x]$$

$$= \min_{\pi_k} \mathbb{E}_k[v_{k+1}((1 + r_{f,k+1})x + \pi_k^\top R_{k+1})], \quad k = 0, 1, \dots, N-1$$

$$v_N(x) = \frac{1}{2} x^2 - (z_0 + \lambda_0)x + \frac{1}{2} z_0^2 + \lambda_0 z_0$$

Denote  $D_k := \mathbb{E}[R_k R_k^\top] = C_k + \mu_k \mu_k^\top$ ,  $k = 1, \dots, N$ .

#### Optimal Portfolio

Consequently, the optimal portfolio is:

$$\pi_k^*(x) = (1 + r_{f,k+1}) \left( \prod_{j=k}^{N-1} (1 + r_{f,j+1}) \right)^{-1} (z_0 + \lambda_0) - x \mathbf{D}_{k+1}^{-1} \mu_{k+1}$$

where the multiplier  $\lambda_0$  is to be determined. By maximizing  $v_0(w_0)$  with respect to the multiplier  $\lambda_0$ , we can derive the multiplier:

$$\lambda_0 = -z_0 + \frac{1}{1 - \prod_{j=0}^{N-1} (1 - \mu_{j+1}^\top \mathbf{D}_{j+1}^{-1} \mu_{j+1})}$$

$$\times \left[ z_0 - \prod_{j=0}^{N-1} (1 + r_{f,j+1}) \right] \left( \prod_{j=0}^{N-1} (1 - \mu_{j+1}^\top \mathbf{D}_{j+1}^{-1} \mu_{j+1}) \right) z_0$$

To summarize, the optimal portfolio is:

$$\pi_k^*(x) = (1 + r_{f,k+1}) \left[ s_k^{-1} \frac{(z_0 - \beta_0 x_0 x_0)}{1 - \beta_0} - x \right] \mathbf{D}_{k+1}^{-1} \mu_{k+1}$$

where  $s_k$  stands for the risk-free growth rate from the beginning of  $k$ -th period to the end, i.e.,  $s_k := \prod_{j=k}^{N-1} (1 + r_{f,j+1})$ , and  $\beta_k := \prod_{j=k}^{N-1} (1 - \mu_{j+1}^\top \mathbf{D}_{j+1}^{-1} \mu_{j+1})$ .

Moreover, the optimal variance is  $2v_0(w_0)$  at the optimal  $\lambda_0$  and at the initial wealth  $w_0$ , i.e., the variance of the terminal wealth under the optimal strategy is:

$$\text{Var}(X_N) = \frac{1}{1 - \beta_0} (z_0 - s_0 w_0)^2$$

The above traces out the efficient frontier as  $z_0$  varies in  $[s_0 w_0, +\infty)$ . The mutual fund theorem still holds.

### 4.8 Continuous-Time Model

#### Market

We can extend the multi-period model to a continuous-time model. Suppose the market has  $m$  risky stocks with prices at time  $t$  denoted as  $S_{i,t}$ ,  $i = 1, \dots, m$ , and a risk-free asset with price  $S_{0,t}$  at time  $t$ . Suppose that the risk-free asset generates net return rate  $rdt$  in an infinitesimally small time period  $[t, t+dt]$ , where  $r$  is a constant, i.e.,

$$dS_{0,t}/S_{0,t} = rdt$$

Suppose that each stock  $i$  generates excess return  $b_i dt + \sum_{j=1}^m \sigma_{i,j} dW_{j,t}$  for some constants  $b_i$  and  $\sigma_{i,j}$ ,  $j = 1, \dots, m$ , where  $(W_{1,t})_{t \geq 0}, \dots, (W_{m,t})_{t \geq 0}$  are  $m$  independent standard Brownian motions. In other words,

$$dS_{i,t}/S_{i,t} = (r + b_i)dt + \sum_{j=1}^m \sigma_{i,j} dW_{j,t}$$

#### Portfolio

Assume that  $\sigma := (\sigma_{i,j})_{i,j=1, \dots, m}$  is invertible. Denote by  $\pi_t = (\pi_{t,1}, \dots, \pi_{t,m})^\top$  the agent's portfolio held in the infinitesimally small time period  $[t, t+dt]$ , i.e.,  $\pi_{t,j}$  dollars are invested in stock  $i$ ,  $i = 1, \dots, m$ , and the remaining is invested in the risk-free asset in this period. Then, the agent's wealth, denoted as  $(X_t)_{t \geq 0}$ , evolves as:

$$dX_t = rX_t dt + \pi_t^\top (bdt + \sigma dW_t)$$

where  $b := (b_1, \dots, b_m)^\top$  and  $W_t := (W_{1,t}, \dots, W_{m,t})^\top$ . Suppose the agent's wealth at the initial time 0 is  $x_0$ .

#### Expected Utility Maximization

Consider maximization of expected utility of wealth at a given terminal time  $T$ :

$$\begin{aligned} & \max_{\pi_t, t \in [0, T]} \mathbb{E}[u(X_T)] \\ \text{s.t. } & dX_t = rX_t dt + \pi_t^\top (bdt + \sigma dW_t) \\ & X_0 = x_0 \end{aligned}$$

where  $u(\cdot)$  is the agent's utility function.

For power utility function  $u(x) = x^{1-\gamma}/(1-\gamma)$ ,  $x > 0$  for certain  $0 < \gamma \neq 1$ , the optimal portfolio is:

$$\pi_t^* = \frac{1}{\gamma} (\sigma \sigma^\top)^{-1} b, \quad t \in [0, T]$$

#### Mean-Variance Analysis

Consider mean-variance portfolio selection. Suppose the target for the expected wealth at the terminal time  $T$  is  $z_0 > x_0 e^{rT}$ . The mean-variance problem is:

$$\begin{aligned} & \max_{\pi_t} (\pi_t)_{t \in [0, T]} \frac{1}{2} \text{Var}(X_T) \\ \text{s.t. } & dX_t = rX_t dt + \pi_t^\top (bdt + \sigma dW_t) \\ & X_0 = x_0 \end{aligned}$$

The optimal portfolio is:

$$\pi_t^* = \left[ e^{-r(T-t)} \frac{z_0 - e^{-b^\top} (\sigma \sigma^\top)^{-1} b T}{1 - e^{-b^\top} (\sigma \sigma^\top)^{-1} b T} - X_t \right] (\sigma \sigma^\top)^{-1} b, \quad t \in [0, T]$$

The optimal variance is:

$$\text{Var}(X_T) = \frac{e^{-b^\top} (\sigma \sigma^\top)^{-1} b T}{1 - e^{-b^\top} (\sigma \sigma^\top)^{-1} b T} (z_0 - e^{rT} x_0)^2$$

## 5 Limit Order Book

### 5.1 Bid, Ask, and Mid Prices

The lowest price among all sell limit orders is referred to as ask price.

The highest price among all buy limit orders is referred to as bid price.

The difference between the bid and ask prices is referred to as bid-ask spread. The average of the bid and ask prices is mid price.

### 5.2 Resolution Parameters

In general, an order  $x = (p_x, \omega_x, t_x)$  submitted at time  $t_x$  with price  $p_x$  and size  $\omega_x > 0$  (respectively,  $\omega_x < 0$ ) is a commitment to sell (respectively, buy) up to  $|\omega_x|$  units of the traded at a price no less than (respectively, no greater than)  $p_x$ .

The lot size  $\sigma$  of an LOB is the smallest amount of the asset that can be traded within it. All orders must arrive with a size  $\omega_k \in \{\pm k\sigma : k = 1, 2, \dots\}$ .

The tick size  $\pi$  of an LOB is the smallest permissible price interval between different orders within it. All orders must arrive with a price that is specified to the accuracy of  $\pi$ .

The lot size  $\sigma$  and tick size  $\pi$  of an LOB are collectively called its resolution parameters.

### 5.3 Limit Order Book

An LOB  $\mathcal{L}(t)$  is the set of all active orders in a market at time  $t$ . The active orders in an LOB  $\mathcal{L}(t)$  can be partitioned into the set of active buy orders  $\mathcal{B}(t)$ , for which  $\omega_k < 0$ , and the set of active sell orders  $\mathcal{A}(t)$ , for which  $\omega_k > 0$ .

### 5.4 Depth

The bid-side depth available at price  $p$  and at time  $t$  is

$$b^p(p, t) := \sum_{x \in \mathcal{B}(t): p_x = p} \omega_x$$

The ask-side depth available at price  $p$  and at time  $t$ , denoted  $a^p(p, t)$ , is defined similarly using  $\mathcal{A}(t)$ .

The bid price at time  $t$  is the highest stated price among all active buy orders at time  $t$ , i.e.,

$$b(t) := \max_{x \in \mathcal{B}(t)} p_x$$

The ask price at time  $t$  is the lowest stated price among all active sell orders at time  $t$ , i.e.,

$$a(t) := \min_{x \in \mathcal{A}(t)} p_x$$

### 5.5 Relative Price

The bid-ask spread at time  $t$  is  $s(t) := a(t) - b(t)$ . The mid price at time  $t$  is  $m(t) := [a(t) + b(t)]/2$ .

For a given price  $p$ , the bid-relative price is  $\delta^b(p) := b(t) - p$  and the ask-relative price is  $\delta^a(p) := p - a(t)$ .

For a given order  $x = (p_x, \omega_x, t_x)$ , the relative price of the order is  $\delta_x := \delta^b(p_x)$  if the order is a buy order, and  $\delta_x := \delta^a(p_x)$  if the order is a sell order.

The bid-side depth available at relative price  $\delta$  and at time  $t$  is

$$b^p(\delta, t) := \sum_{x \in \mathcal{B}(t): \delta_x = \delta} \omega_x$$

The ask-side is defined similarly.

### 5.6 Price changes in LOBs

Consider a buy (